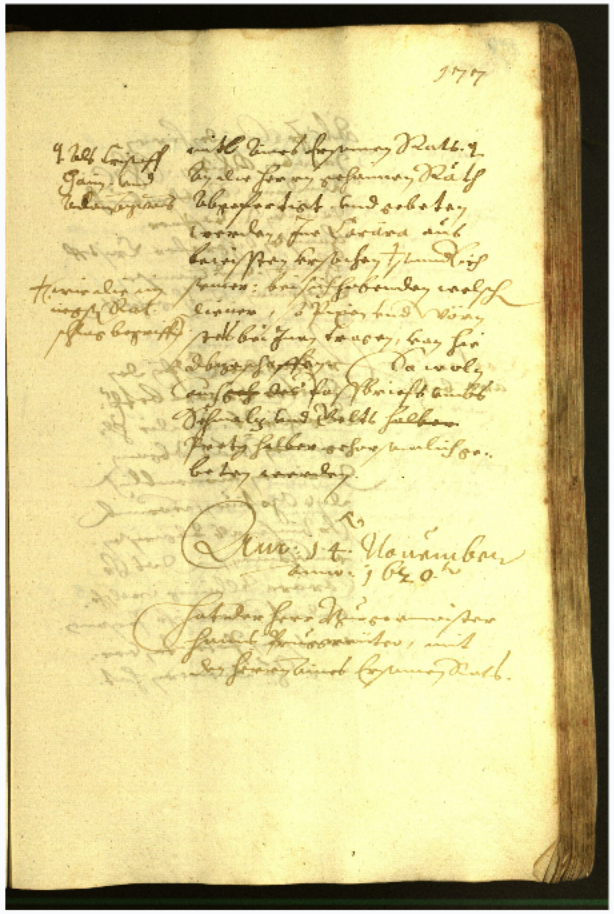
FYP RESEARCH PROGRESS · 27 SEPTEMBER 2026

Active learning for historical OCR

Can better page selection reduce the labelled data needed to adapt small vision-language models?

Evidence through EXP-013 · READ2016 / Bozen

Dataset and the unit of annotation



Historical German handwriting

Official split	Pages	Text lines
TRAIN pool	350	8,367
VAL	50	1,043

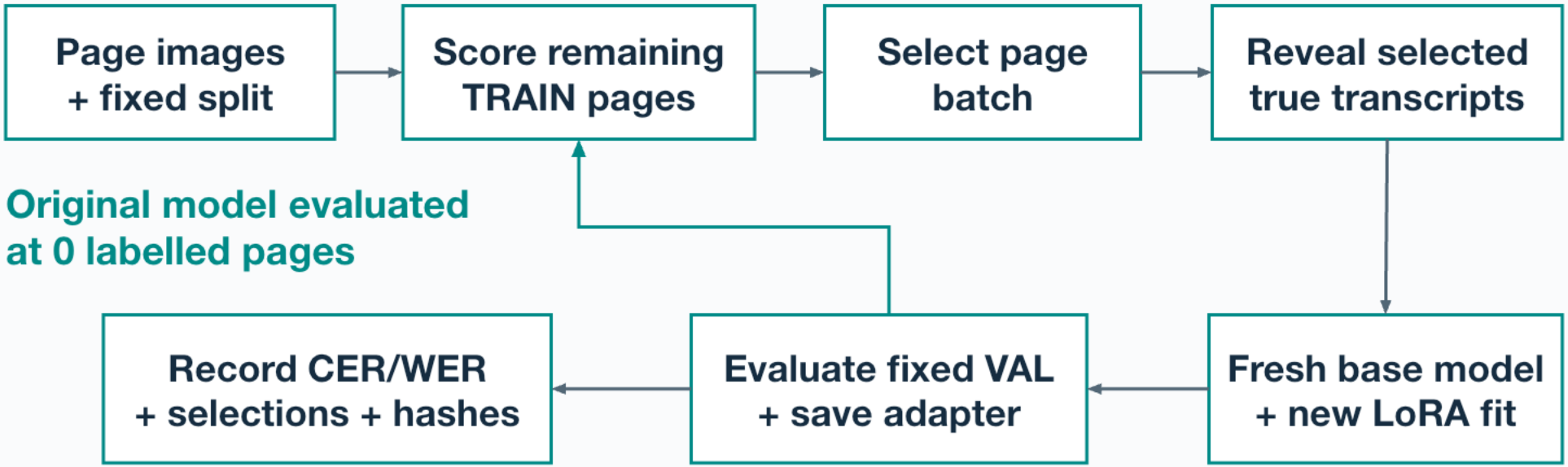
Select one TRAIN page → reveal every line transcript

Existing true labels simulate annotation.
Model scoring sees images; VAL truth stays with the evaluator.

8 VAL pages for recipe checks · 42 fixed VAL pages for reported curves

Source: READ2016 / Bozen v1.2.0 · Toselli et al. · CC BY 4.0

The active-learning pipeline



Each round fits all acquired labels; later uncertainty uses that arm's preceding adapter.

TRAIN oracle ≠ VAL evaluator · Random skips scoring · Stop after 3 fixed rounds

Early checks separated engineering from OCR quality

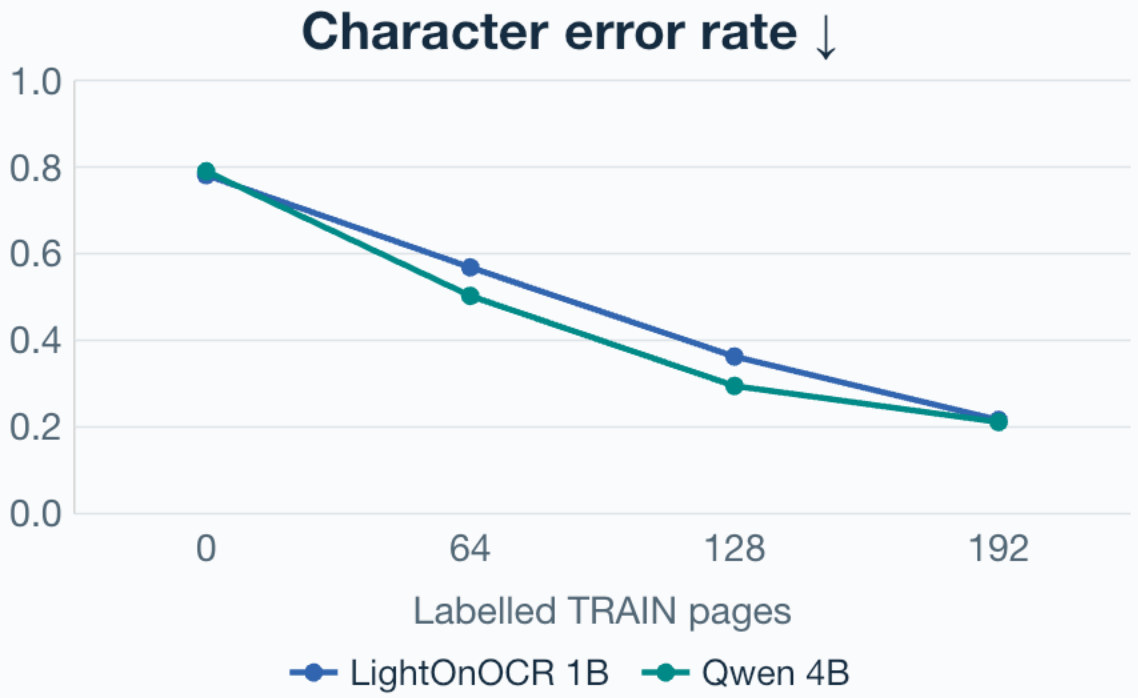
Experiments / question	Observed result	Decision
001–002 Can remote execution complete?	Timeout; then malformed/truncated OCR and no accepted post-fit checkpoint	Repair runtime and checkpoint acceptance
003–005 Does the full round work?	Fit → reload → evaluate worked CER ≈ 1.0; line-box F1 = 0	Engineering passed; OCR quality did not
006–008 Can the joint recipe be improved?	More pages, stronger fitting, compact rows Still invalid/repetitive or capped outputs	Pause this recipe; isolate transcription

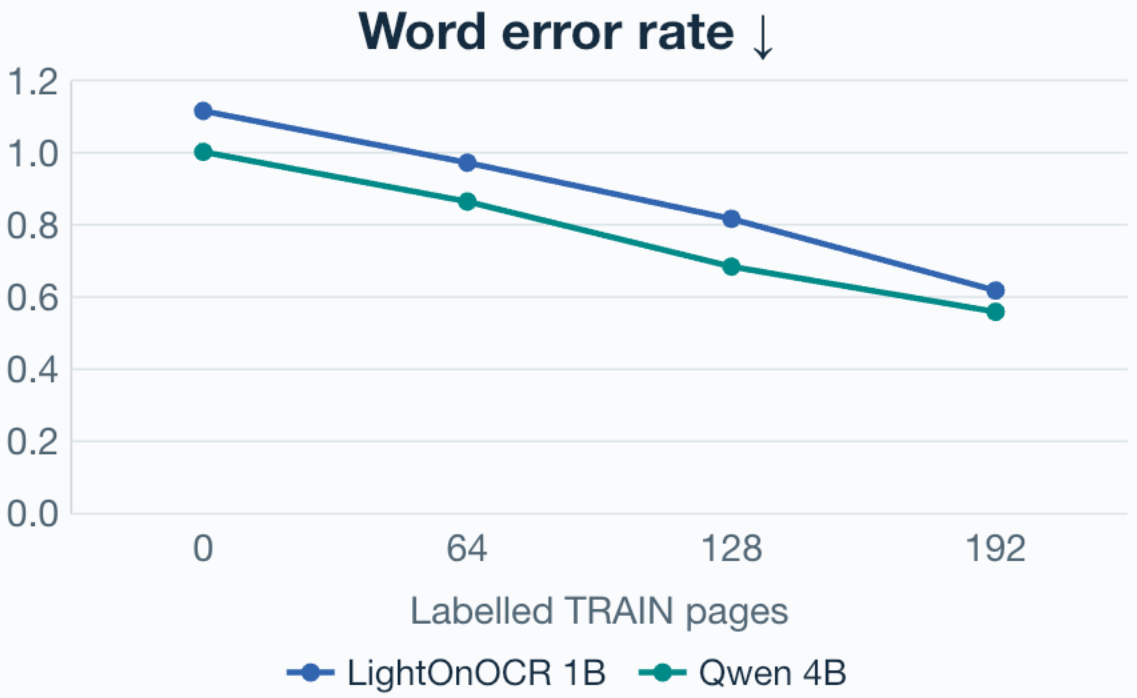
Why change the task?

An unusable OCR output makes selection effects hard to interpret.
Test recognition first, then investigate which labels are useful.

Sources: results/EXP-001-002.md · EXP-003-005.md · EXP-006-008.md

Both models learn whole-page transcription





Same page budgets · 3 epochs per fresh fit · 42 fixed VAL pages

At 192 pages: CER 0.216 / 0.211 · Qwen capped pages 10 → 3 → 1 → 0

CER = character edits / reference characters; WER = word edits / reference words. Lower is better.

Sources: EXP-009 VAL42 pilot · EXP-010 VAL42 · shared page-text recipe

Whole-page selectors show no consistent advantage

0 pages
Original model

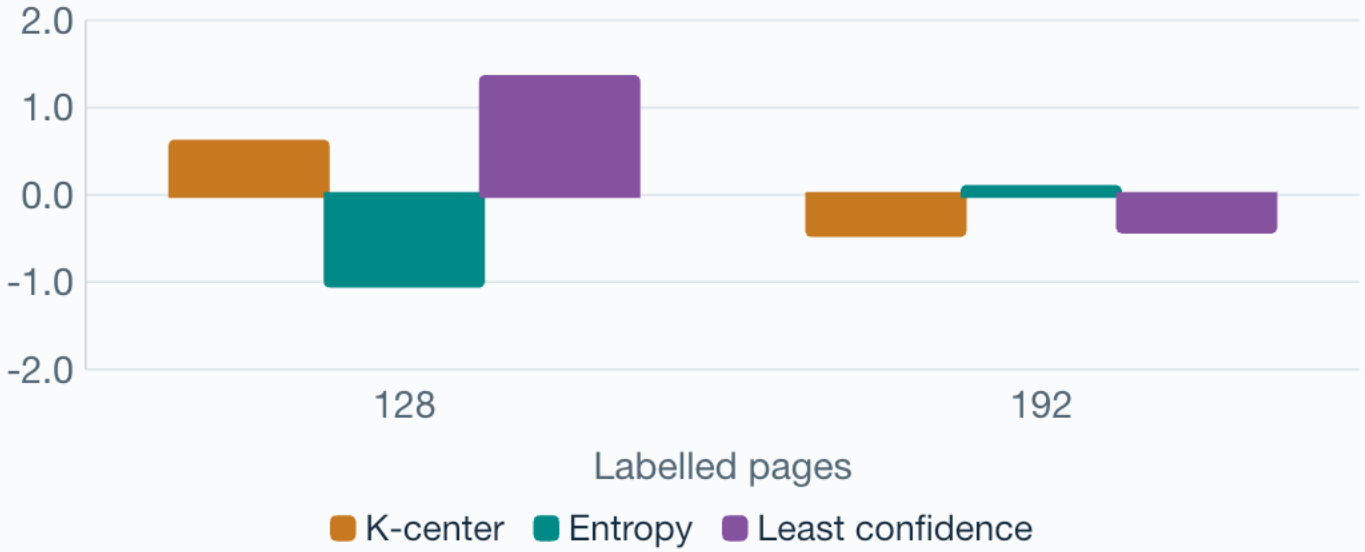
→

64 random pages
Shared checkpoint

→

128 → 192 pages
Strategies diverge

Mean paired ΔCER (percentage points)



What repeated seeds changed

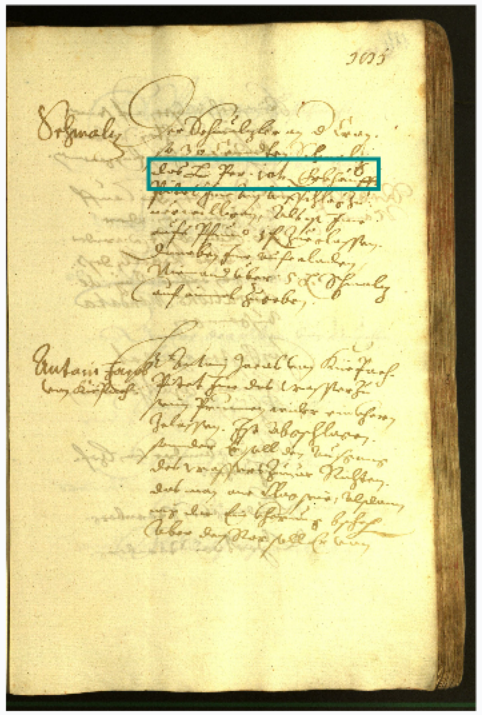
Seed 824 looked promising.
The other seeds changed the ranking.

Output caps affected small gaps.
No method led consistently at both budgets.

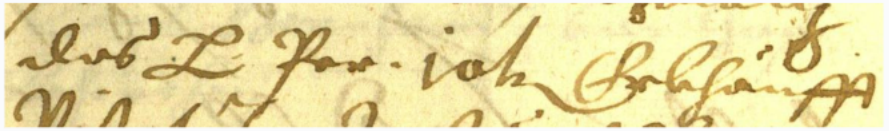
Means across 3 paired seeds; not significance estimates. Full VAL42 scores remain primary.

Source: results/EXP-011.md · negative Δ favours the selector; positive Δ favours random

Known-box recognition removes a source of ambiguity



Whole page



Supplied line box → Qwen transcribes only this crop

Short text outputs in source reading order

Why this helps

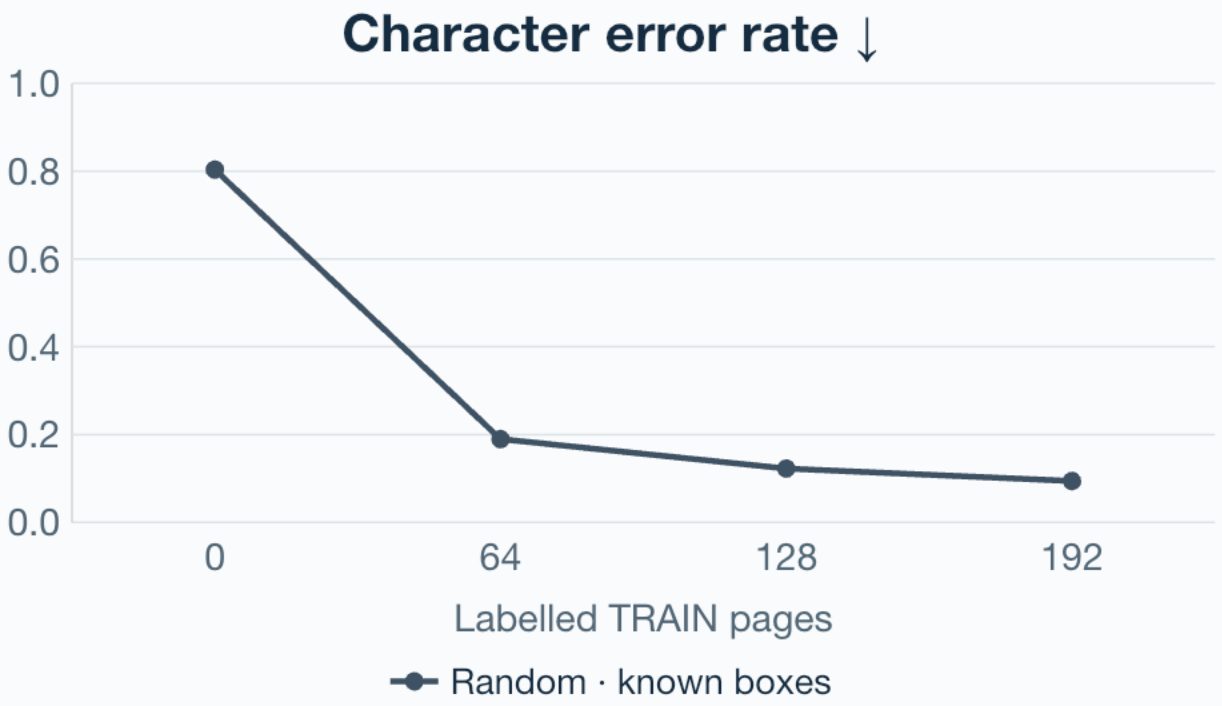
Separate recognition from localization.

Reduce long-output failures.

Study selection with usable OCR.

This is a different task and recipe; its CER is not a fair head-to-head gain over whole-page OCR.

Random selection is a strong known-box baseline



Pages	Train lines	CER	WER
0	0	0.804	1.087
64	1,537	0.190	0.581
128	3,037	0.123	0.452
192	4,567	0.094	0.370

The largest drop comes with the first 64 pages

1 epoch per fresh fit · 3 fixed rounds · capped VAL regions: 2 → 0 → 0 → 0

Establishes adaptation and a control curve; no model-informed selection comparison yet.

Source: results/EXP-012.md · same 42 VAL pages / 861 lines at every checkpoint

Selection starts with each strategy's own first batch

Strategy	How it chooses pages
Random	Seeded random pages from the remaining pool
K-center	Frozen visual coverage; centroid-nearest first, then farthest
Entropy	Highest mean token-normalized uncertainty
Least confidence	Lowest mean geometric token confidence



No shared random warm start · Later uncertainty uses that arm's previous adapter

0-page VAL42 baseline reused from EXP-012 after identity checks and local metric recomputation

Uncertainty leads at 24 pages; random leads later



At 24: entropy improves CER vs random by 5.45 percentage points

At 72: random CER 0.175 · entropy 0.178 · k-center 0.184 · confidence 0.188

0 capped VAL outputs after training · entropy/confidence share 37 of 72 pages

Same VAL42 / 861 lines · lower is better · original-model baseline reused from EXP-012

Current findings and the next research test

Established

Domain fine-tuning improves recognition.

Both page and crop pipelines run.

Random is a necessary strong control.

Still unproven

Repeatable annotation savings.

Independent-document generalization.

Box detection or human-time savings.

EXP-013: one seed; LoRA initialization was not explicitly seeded. Official validation only.

Next: repeat the 24/48/72 comparison with controlled LoRA initialization

Seed before constructing adapters; pair multiple seeds; retain random and fixed evaluation.
Then test whether OCR uncertainty plus diversity improves early-budget selection.

Methodological context: Munjal et al., CVPR 2022 · historical recognition: Aradillas et al., 2020

References and experiment records

Sener & Savarese · ICLR 2018

Core-set coverage motivates k-center; classification ≠ OCR.

Munjal et al. · CVPR 2022

Strong random controls and stochasticity-sensitive evaluation.

Aradillas, Murillo-Fuentes & Olmos · 2020 preprint

Few-label adaptation for historical handwritten recognition.

Bai et al. · Qwen3-VL Technical Report, 2025

Open general VLM family; Qwen3-VL-4B-Instruct used here.

Taghadouini, Cavaillès & Aubertin · LightOnOCR, 2026

Compact page-transcription VLM; 1B base used in EXP-009.

Toselli et al. · READ2016 / Bozen v1.2.0

Page images + source line annotations; CC BY 4.0.

Experiment reports: EXP-001–002 · 003–005 · 006–008 · 009 · 010 · 011 · 012 · 013